



The effect of rearing environment on feather pecking in young and adult laying hens



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ABSTRACT

Although the rearing period has an important influence on the development of feather pecking in laying hens, few studies have quantified the risk factors operating on commercial farms during this time and identified their long-term impact. Our aim was to conduct a longitudinal study to investigate the effect of rearing environment on feather pecking in young and adult laying hens. Thirty-four flocks from 29 rearing farms were recruited and visited at the beginning, middle and end of the rearing period and once at lay (35 weeks). Twelve flocks were beak trimmed. Information on rearing environment was used to create models predicting feather pecking and plumage damage during rear and lay, using the multilevel statistical software MLwiN 2.25. Across all flocks, gentle feather pecking (GFP) was observed during 94% of the visits at both rear and lay, at 1.3 and 1.0 bouts/bird/h respectively. Severe feather pecking (SFP) was observed during 27% of the visits during rear and during 65% of the visits at lay, with a mean rate of 0.4 pecks/bird/h during rear and 1.9 pecks/bird/h at lay, across all flocks. The mean percentage of the flock with missing feathers was 12% at 16 weeks and 49% at lay. The mean individual feather score at lay was 21 (range 6–24 (best)). The study confirmed that feather pecking and feather damage occur during the rearing period. Statistical modelling further showed that the percentage of the flock with missing feathers was significantly lower and individual feather scores significantly higher (better) at lay, in flocks where feather pecking had not started at the end of rear. The three models on the effect of rearing environment on GFP, SFP and the percentage of the flock with missing feathers during rear contained 21 significant variables. Approximately a third of those related to house climate (temperature, humidity, sound, light and dust levels), while another third related to foraging. Foraging itself appeared in all three models, confirming that good foraging is one of the major factors in reducing feather pecking. The four models on the effect of rearing environment on GFP, SFP, the percentage of the flock with missing feathers and individual feather scores at lay contained 17 significant variables and sound level was significant in three of the four. The analysis further indicated that experienced rearing staff was protective against feather pecking at both rear and lay and that feather pecking increased with an increasing number of diet changes during rear.

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1. Introduction

Feather pecking is the (non-aggressive) pecking at or pulling out of the feathers of other individuals and it remains the most prevalent behavioural problem in the laying hen sector. In a European project on layer welfare, data collected from a number of studies indicated that feather pecking was observed in 40–80% of flocks in

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non-cage systems (Blokhuis et al., 2007). The damage occurring as a result of feather pecking is lower in cages because the group size is smaller (fewer available victims) and it is easier to locate and remove the pecker. In 2012 however, the conventional cage system was banned throughout the EU and in some EU-countries beak trimming is (Switzerland, Sweden, Norway and Finland), or will be (in the UK the date is currently set for 2016), prohibited as well. These bans are likely to increase feather damage caused by feather pecking. Apart from welfare problems, feather pecking can result in economic loss by increasing mortality and thereby reducing production and by worsening the feed conversion ratio (Tauson and Svensson, 1980). Feather pecking is a multi-factorial problem which can be influenced by genetic background, early life history and environmental factors (Rodenburg et al., 2013). Factors such as litter availability (Huber-Eicher and Wechsler, 1997; Huber-Eicher and Sebö, 2001; Bestman et al., 2009), stocking density (Huber-Eicher and Audigé, 1999; Bestman et al., 2009), perches (Huber-Eicher and Audigé, 1999; Knierim et al., 2008), nutrition (van Krimpen et al., 2009), feeders and drinkers (Knierim et al., 2008; Drake et al., 2010), house climate and lighting (Drake et al., 2010; Lambton, 2008; Bestman et al., 2009) are all known to affect feather pecking. It is therefore hard for farmers to identify the specific causes affecting their flock. Farmers are advised to optimise housing and management but despite their best efforts, the problem may still occur. There are several ways to reduce problems caused by feather pecking (e.g. beak trimming, reducing the light intensity), but these treat the symptoms rather than addressing the causes (Nicol et al., 2013). It proves very hard to stop feather pecking once it has started and the focus should therefore be on prevention.

Recent studies suggest that the influence of the rearing period is more important than previously thought. Knierim et al. (2008) analysed risk factors leading to feather damage during lay in 23 organic rearing farms and found that 79% of the variation in feather damage could be ascribed to rearing conditions. Lambton et al. (2010) and Drake et al. (2010) found that SFP was higher and plumage damage occurred earlier, when feather pecking was already being performed by the pullets on arrival at the laying farm. Similarly, Bestman et al. (2009) found that 54% of 24 organic flocks showed feather damage during rear and 90% of these also showed it during lay, while of the remaining 46% of flocks that did not show feather damage during rear, 71% continued with good plumage during lay. Several other studies have identified a high rearing prevalence of feather pecking, including Huber-Eicher and Sebö (2001) who found that 77% of 25 flocks had already developed significant amounts of feather pecking (>30 pecks/30 birds/30 min) at 14 weeks of age. Lambton (2008) further found that GFP was already seen in 100% and SFP in 47% of 15 flocks at the age of 10 weeks. These figures suggest that feather pecking can frequently develop early in the life of the birds, though it may not be very apparent, as young birds moult several times so plumage damage is usually subtle.

The effect of rearing environment on feather pecking has been studied in several controlled experiments, which are useful, but cannot easily replicate the size and complexity of commercial environments. Only six epidemiological

studies have investigated the effect of rearing environment on feather pecking on commercial farms (Gunnarsson et al., 1999; Huber-Eicher and Audigé, 1999; Knierim et al., 2008; Lambton, 2008; Bestman et al., 2009; Drake et al., 2010). As feather pecking appears to develop early on in the life of the birds, it is important to start observations as early as possible. However, only two of the epidemiological studies looked at the birds during the entire rearing period (Lambton, 2008; Bestman et al., 2009) and the sample size of Lambton (2008) was relatively small (15 flocks). Knierim et al. (2008) and Drake et al. (2010) observed the birds at the end of rear only and the two remaining studies used a retrospective approach or a survey. The current longitudinal study of birds aged between 1 and 35 weeks aimed to quantify the management risk factors for feather pecking that occurred during both the rearing and subsequent laying period.

2. Materials and methods

Thirty-four flocks, from 29 rearing farms located in England, Wales and Scotland were recruited. Each flock was visited at approximately 1, 8 and 16 weeks of age, corresponding with the beginning, middle and end of the rearing period. The flocks were further visited once at lay, at approximately 35 weeks of age. During these visits information on the farms, houses, flocks and management was collected. This information was turned into 180 variables, or risk factors, which were used to derive statistical models predicting four outcome variables: GFP, SFP, the percentage of the flock with missing feathers and individual feather scores. Information on both direct feather pecking and the indirect results of it, feather damage, was gathered because feather pecking may not always be observed while feather damage is always there to see. As young birds usually do not show much feather damage (even if feather pecking occurs), missing feathers were scored only at the end of the rearing period and at 35 weeks and individual feather scores were measured only at 35 weeks.

2.1. Animals and housing

The 34 flocks comprised the following breeds: Columbian Blacktail (18), Lohmann Brown (6), Hy-Line (4), Goldline (2), MS (1), Blackrock (1), Brown Nick (1) and Shaver (1). Twelve flocks were beak trimmed (infrared) and the rest had intact beaks. The average flock size was 6548 (range 282–38,822) during rear and 4374 (276–15,848) at 35 weeks. Sixteen flocks were organic, 17 were barn and one was free range during rear. At lay, all organic flocks remained organic and all the barn flocks became free range apart from one, which remained barn.

The rearing period lasted on average 16 weeks, roughly divided into the brooding period (first 6–8 weeks, where chicks received supplementary heat) and the rearing period. Twenty-four of the flocks stayed in the same house during this entire period. Four flocks were brooded in one house and then split up over several (similar) houses. The last six flocks were moved to a new house, which resembled a laying house (litter and slats), after brooding. Another option was to integrate the slatted area in the brooding

house and open it up after about 6–8 weeks, which happened for six of the 24 flocks that stayed in the same house for the entire rearing period. Seven flocks were reared in a single-tier system and one in a multi-tier system, the rest were housed in a floor or a floor and slats system. At lay, 19 of the 34 flocks went to farms different from those where they were reared. During the laying period all flocks were housed in a floor and slats system, apart from two flocks which were housed in a multi-tier system and one flock which had a floor system. Management (temperature, light, feeding, and vaccinations) of all flocks was according to standard farm practices. Barn and free range flocks were housed according to RSPCA Freedom Foods standards, while the organic flocks followed Soil Association, Organic Farmers and Growers or Organic Food Federation standards.

2.2. Data collection

Farm visits took place between March 2009 and June 2011. Each flock was visited four times in total, apart from three flocks which were not seen at 16 weeks, resulting in $4 \times 34 - 3 = 133$ farm visits. Table 1 shows which activities were carried out during each visit. General information about the farm, management and the flock was gathered by way of detailed interviews with the farmer. The first questions concerned the farm and its environment and any previous history with feather pecking. The next group of questions related to the flock that was going to be part of the study and its house, health and management (lighting, climate control, enrichment, inspection information and diet). The questionnaires were adapted from those used by Lambton et al. (2010).

Length, width, minimum and maximum ceiling height of the house were measured with a laser range finder (Bosch, DLE 50 Professional). If birds had access to one or several tiers of slats this was included in available floor space. Feeder and drinker type were recorded and available feed and drink space per bird was measured. Any perches available to the birds were measured, as well as their height from the floor and the height between the perches.

During all visits house climate was recorded at up to six (depending on house size) equally distributed locations in the house. This way the average climate within the house within the visit was calculated, as well as the highest, lowest and the range between the highest and lowest readings. All measurements were taken at bird head height. Temperature (°C) and relative humidity (%) were recorded by means of a pocket weather meter (Kestral 4000). Sound intensity was recorded by means of a digital sound level meter with a range of 40–130 dB (CEM, DT-85A). Light intensity was recorded by means of a light meter (RS, 180-7133) with a range of 0–50,000 lx and with light type selection option (tungsten, fluorescent, day light and mercury). Lastly, ammonia levels were recorded by means of an ammonia meter (Industrial scientific, GasBadge® Pro) with a range of 0–100 ppm. A qualitative assessment was made of the air quality in the house by scoring the difficulty to breathe (easy/average/difficult) and the level of dust (low/average/high) once. The litter quality was assessed by

estimating the percentage of caked (non-friable) and wet litter.

During each visit feather pecking was recorded for 10 min in four different areas of the house (total 40 min) to obtain a representative sample of the entire flock. The observation areas were chosen in such a way that each contained a combination of the litter area, the slatted area and the perches, depending on whether these were available to the birds. This was done to account for the possibility that feather pecking could occur differentially in different areas. GFP was classified as mild, repetitive pecking at the feathers of another bird and SFP as plucking at or pulling out the feathers of another bird. For the observations an area measuring approximately 1 m², containing birds which were clearly visible and not dust bathing, was selected. Before and after each observation period the number of birds present in the observation area was counted to calculate the average number present at the time of observation. Stocking density was taken into account later on, as it was one of the explanatory variables tested in the models. Explorative gentle feather pecks (aimed at particles on the feathers rather than at the feathers themselves) were not recorded. Each peck or each bout (in case of GFP) was scored as one occurrence, with one bout ending when pecking stopped for over 4 s or when pecks were directed at another body part. Because Lambton et al. (2010) found a correlation between feather pecking and foraging, the latter was scored as well. This was done by means of 10 scan samples at 1 min intervals during each of the four 10 min observation periods. After each scan the number of foraging birds within the observation area was counted. Foraging was defined as pecking and scratching at the litter or moving with the head in a lower position than the rump.

Feather condition was assessed on an individual and a flock level. Individual feather scores were recorded at 35 weeks for 20 birds per flock in different areas of the house using the method developed by Tauson et al. (2005). With this method the bird's body was divided into six parts (neck, breast, vent, back, wings and tail feathers) and each part was scored with a score of 4 (perfect), 3 (scruffy), 2 (bald patches) or 1 (bald), resulting in a total score of 6–24. In addition, comb and vent condition were scored as follows: 4 (no wounds), 3 (<5 pecks or scratches), 2 (5 or more pecks or scratches) and 1 (bloody wounds). For 220 birds from the 12 beak trimmed flocks the condition of the beak was also scored using the following scales: 4 (no abnormalities), 3 (lower half longer), 2 (lower half longer and not straight) and 1 (lower half longer and split). The variation in comb, vent and beak condition turned out not to be great enough for statistical analysis. The feather condition of each flock was further assessed by estimating the percentage of birds with missing feathers at 16 and 35 weeks by scoring whether or not the birds had missing feathers on the lower back, in 100 animals in different areas of the house (Bestman et al., 2009).

2.3. Analysis

Models predicting each of the four outcome variables were tested in MLwiN 2.25 (Rasbash et al., 2009). This was done separately for the rearing and the laying period,

Table 1

Description of which activities were done during each farm visit.

	Visit 1	Visit 2	Visit 3	Visit 4
Age birds ^a	1 week	8 weeks	16 weeks	35 weeks
Interview farmer	X	X	X	X
Measuring house	X	X (if changed)		X
Measuring house climate	X		X	X
Feather pecking observations	X	X	X	X
% of flock with missing feathers			X	X
Individual feather scores				X

^a Due to logistical reasons not all visits took place at the exact ages stated here.

resulting in seven models (as individual feather scores were only measured at 35 weeks). Depending on whether the outcome variables were measured during only one or more visits (in which case the model needed to account for repeated measures), two or three level models were created ((visits within) flocks within farms). As SFP only occurred in less than a third of the visits during rear it was coded as a binary variable, to test, using logistic regression, whether the predictor variables had an influence on its presence/absence rather than on its rate. Further to this, an analysis assuming normal errors of the rate of SFP in only the flocks that performed the behaviour was performed to address the question of what affected the rate of SFP within flocks that displayed SFP. This was done for data from 35 weeks only, because the model for the probability of SFP only gave one significant variable.

Before entering the variables in the model a reference category was set for categorical variables, while continuous variables were centred around their grand mean. Each model contained a constant term. GFP during rear and lay were log transformed after adding 0.01 to avoid zero values and SFP at 35 weeks was natural log transformed (excluding the binary analysis). The percentage of the flock with missing feathers at 35 weeks was transformed by changing the percentages to proportions and then taking their square root followed by their $\arcsin(\sin^{-1})$ according to the following formula: $\arcsin((\text{percent missing feathers}/100)^{0.5})$. Given the large number of potential explanatory variables an initial sifting was carried out. Each potential variable was fitted as the only predictor and variables that did not reach a significance of 0.05 were dropped. It is more usual to use a significance level of 0.10 for this initial screening, however, a more stringent level was used given the large number of potential variables. The thus selected variables were then grouped into approximately similar categories (e.g. all predictors related to perching) and stepwise forward regression was performed on each of the groups to identify which of them remained significant after being submitted to stepwise regression together. Stepwise forward regression was then performed on the retained variables in a similar way. Once this process was completed, a final check was done by once more adding in the remaining predictor variables to the model one at a time. The models were tested for the effects of age, as well as for possible quadratic relations in the continuous variables. To test for a cyclic seasonal effect, hatch and visit date were also entered within each model: all dates were transformed into days from 1 to 365 and sine and cosine transformed. Entering these two variables into a model allowed

testing for a sinusoidal pattern ($\sin(2\pi(\text{day of year}/365))$ and $\cos(2\pi(\text{day of year}/365))$).

3. Results

The estimates for each of the variables retained in the models are reported in [Tables 2–7](#). None of the outcome variables was significantly influenced by stocking density. Positive estimates indicate that an increase in the variable was associated with an increase in the outcome variable and vice versa for negative estimates. The mathematical models based on these estimates can then be used to calculate predicted values for the outcome variables of the models based on the levels of the predictor variables. For instance, the predicted value for the rate of GFP for a highest house temperature of 23 °C, when the other variables were fixed at their default values would be ([Table 2](#)): $\text{Log GFP} = -0.014 + (-0.026 \times 23 - 21.8) + (-0.005 \times 23 - 21.8 \times 23 - 21.8) = -0.038$ bouts/bird/40 min. Keeping in mind that these data were log transformed after adding 0.01, the actual rate would be: $10^{-0.038} - 0.01 = 0.91$ bouts/bird/40 min.

3.1. Gentle feather pecking

At one week old, GFP was seen in 94% of the 34 flocks. The mean rate was 0.8 bouts/bird/h for all flocks, or 0.9 bouts/bird/h for the 32 flocks that showed GFP. The highest observed rate was 2.6 bouts/bird/h. At eight weeks, GFP was observed in all but one of the flocks (97%). The mean rate was 1.4 bouts/bird/h, while the highest observed rate was 10.9 bouts/bird/h. At 16 weeks, GFP was seen in 90% of the 31 flocks (as three third visits were cancelled). The mean rate was 1.8 bouts/bird/h for all flocks, or 2.0 bouts/bird/h for the 29 flocks that showed it. The highest observed rate was 8.5 bouts/bird/h. The analysis on the effect of rearing environment on GFP during rear resulted in 10 significant variables ([Table 2](#)). [Table 2](#) shows the estimates for each of the significant variables plus their standard errors, as well as the mean and range for continuous variables and the categories for categorical variables. GFP during rear was reduced when the highest house temperature was above or below 20 °C, when there was a larger variation of sound and light level in the house (within the visit), with less variation in humidity levels, low dust levels, in larger flocks, in houses with both nipple and bell drinkers and in non-tiered systems. GFP was further reduced when a higher percentage of birds were observed to be foraging during the 40 min observation

Table 2

Risk factors for the effect of rearing environment on the rate of Log (gentle feather pecking + 0.01) at rear. The table shows the mean and range of continuous variables and the different categories for categorical variables.

Variable	Mean + range/categories	Est (SE)	P-value	N ^b
Constant		−0.014 (0.103)	0.45	92
Highest house temperature (°C)	21.8 (10.1–30.6)	−0.026 (0.008)	0.0006	92
Highest house temperature ² (°C) ^a		−0.005 (0.001)	<0.0001	92
Range sound levels (dB)	9.5 (0–54)	−0.014 (0.004)	0.0002	92
Range humidity (%)	11.8 (0–34.9)	0.010 (0.004)	0.006	92
Range light levels (lux)	48.7 (0–1703)	−0.00092 (0.00020)	<0.0001	92
Foraging (% of observed birds)	18.1 (0–60.5)	−0.011 (0.003)	0.0001	92
Flock size (birds)	6560 (94–38,822)	−0.00002 (0.00001)	0.02	92
Breed	Lohmann brown	−0.051 (0.106)	0.32	15
	Hy-Line	−0.005 (0.108)	0.48	11
	Goldline	−0.013 (0.135)	0.46	6
	Blackrock	−0.262 (0.233)	0.13	2
	Brown nick	−0.866 (0.220)	<0.0001	2
	MS	−0.179 (0.183)	0.16	3
	Shaver	0.882 (0.238)	0.0001	2
	CBT	Reference category		51
Dust level in house	Dusty	0.757 (0.148)	<0.0001	7
	Average	0.007 (0.073)	0.46	38
	No dust	Reference category		47
Drinker type	Bell	0.188 (0.090)	0.02	10
	Bell & Nipple	−0.262 (0.109)	0.008	22
	Nipple	Reference category		60
Tiered system	No	−0.222 (0.096)	0.01	70
	Yes	Reference category		22

^a This is a squared term representing the significance of a quadratic (curved) relationship. Climatic variables are averages from within the house within one visit.

^b The 92 refers to the total sample size of 99, minus the number of missing values.

Table 3

The effect of rearing environment on the rate of Log (gentle feather pecking + 0.01) at lay. The table shows the mean and range of continuous variables and the different categories for categorical variables.

Variable	Mean + range/categories	Est (SE)	P-value	N
Constant		1.006 (0.222)	<0.0001	32
Highest sound level at 16 weeks (dB) ^a	62.2 (32–72)	0.059 (0.010)	<0.0001	32
Number of diet changes at rear	1.7 (1–3)	0.418 (0.131)	0.0007	32
Age (days)	225.5 (238–336)	−0.009 (0.003)	0.001	32
Sin visit date	0.41 (−0.82–1.0)	0.325 (0.152)	0.02	32
Cos visit date	0.02 (−0.99–0.98)	−0.194 (0.141)	0.08	32
Chick condition at placement (1 day old)	Good	−1.214 (0.215)	<0.0001	28
	Poor	Reference category		4
Beak trimmed	No	−0.890 (0.143)	<0.0001	20
	Yes	Reference category		12

^a Highest sound level is averaged from within the house within one visit.

periods. The amount of GFP was also associated with breed, it was higher in Shavers (0.882 (0.238), $P=0.0001$) and lower in Brown Nicks (−0.866 (0.220), $P<0.0001$) compared with CBT's, though this result should be treated

with great caution as both breeds were represented by only one flock. As the models are quite complex, specific effects (for instance the effect of highest house temperature on GFP) can only be seen after plotting the predicted

Table 4

The effect of rearing environment on the probability of severe feather pecking at rear. Parameter estimates from the binary logistic regression. The table shows the mean and range of continuous variables and the different categories for categorical variables.

Variable	Mean + range/categories	Est (SE)	P-value	N
Constant		−2.675 (1.102)	0.008	99
Light (h/day)	14.8 (9.8–24)	−0.267 (0.088)	0.001	99
Experience chicks' main caretaker (years)	11.9 (0–40)	−0.117 (0.045)	0.005	99
Foraging (% of observed birds/40 min)	18.1 (0–60.5)	−0.125 (0.040)	0.0009	99
Lowest ceiling height (m)	2.0 (0.7–3.1)	−1.706 (0.728)	0.009	99
Number of diet changes at rear	1.7 (1–3)	4.131 (1.196)	0.0003	99
Farm status	Free range	−5.194 (2.246)	0.01	3
	Barn	−2.921 (1.349)	0.02	43
	Organic	Reference category		53
Beak trimmed	No	2.678 (1.289)	0.02	65
	Yes	Reference category		34

Table 5

The effect of rearing environment on the probability (top) of severe feather pecking and on the rate (bottom) of Ln (severe feather pecking + 0.01) at lay. The table shows the mean and range of continuous variables and the different categories for categorical variables.

Variable		Mean + range/categories	Est (SE)	P-value	N
Probability	Constant		0.738 (0.407)	0.04	34
	Range sound at 16 weeks (dB) ^a	7.8 (0–18)	0.203 (0.098)	0.02	34
Rate	Constant		0.196 (0.167)	0.12	22
	Age	257.7 (238–336)	0.020 (0.005)	<0.0001	22
	Sin visit date	0.35 (–0.82 to 0.99)	0.031 (0.266)	0.45	22
	Cos visit date	0.13 (–0.99 to 0.98)	0.481 (0.214)	0.01	22
	Feeder type	Pan	–0.542 (0.365)	0.07	4
		Compartmentalised pan	1.133 (0.352)	0.0006	4
		Mix	–1.211 (0.437)	0.003	2
		Chain	Reference category		12

^a Range sound level is averaged from within the house within one visit.

Table 6

The effect of rearing environment on the percentage of the flock with missing feathers at 16 weeks (top) and at lay (bottom) (ASIN((percent missing feathers/100)^{0.5})). The table shows the mean and range of continuous variables and the different categories for categorical variables.

Variable		Mean + range/categories	Est (SE)	P-value	N
16 weeks	Constant		9.833 (1.549)	<0.0001	30
	Severe feather pecks/bird/40 min	0.28 (0–3.58)	6.838 (2.403)	0.002	30
	Percentage litter area (% of house)	82.7 (40–100)	–0.197 (0.069)	0.002	30
	Foraging (%)	10.4 (0–54.8)	–0.290 (0.131)	0.01	30
Lay	Constant		1.162 (0.135)	<0.0001	32
	Average sound level at 16 weeks (dB) ^a	58.3 (32–66)	0.021 (0.008)	0.004	32
	Number of people inspecting at rear	2.2 (1–6)	–0.213 (0.074)	0.002	32
	Age (days)	254.9 (238–336)	0.009 (0.003)	0.001	32
	Feather pecking started at 16 weeks	No	–0.457 (0.138)	0.0005	27
		Yes	Reference category		5

^a Average sound level is averaged from within the house within one visit.

graphs, which have not been included here due to space limitations.

At 35 weeks GFP was observed in 32 of the flocks, while it was not observed in two flocks. The mean rate was 1.0 bouts/bird/h for all 34 flocks combined or 1.1 bouts/bird/h for the 32 flocks that performed GFP. The rate was slightly lower than during the rearing period. The highest observed rate was 3.7 bouts/bird/h, which was less than half of the highest rate seen during rear. The analysis for the effect of rearing environment on GFP at 35 weeks resulted in six significant variables (Table 3). The rate of GFP at 35 weeks was reduced with lower maximal sound levels in the house at the end of rear, good condition of chicks at placement, no beak trimming, higher age and fewer diet changes during rear. GFP also varied with visit date, it was lowest in winter. In two flocks GFP was not observed at all. The inclusion of these two flocks in the statistical model provided unsatisfactory residuals. Satisfactory residuals are required to meet the assumptions of the statistical modelling. When the model was run with the two outliers included, 'age' and 'number of diet changes' became insignificant. For those

reasons these two flocks were excluded from the analysis and so the results presented in Table 3 are only relevant with regard to flocks in which GFP was observed across the range 0.2–3.7 pecks/bird/h.

3.2. Severe feather pecking

During the rearing period, SFP was observed in 21 of the 34 flocks during 27 of the 99 visits only. Fourteen flocks showed it at 16 weeks and in five of those the behaviour was also seen at earlier ages, one flock showed it on all three occasions. The mean rate of SFP was 0.4 pecks/bird/h for all 99 visits combined or 1.4 pecks/bird/h for the 27 visits where it occurred. More precisely, SFP was observed in 6% of the 34 flocks at one week, with a mean rate of 0.1 pecks/bird/h for all flocks or 0.7 pecks/bird/h for the two flocks that showed SFP. The highest observed rate was 1.4 pecks/bird/h. At eight weeks, SFP was seen in 32% of the 34 flocks with a mean rate of 0.7 pecks/bird/h for all flocks or 2.1 pecks/bird/h for the 11 flocks that showed the behaviour. The highest observed rate was 7.6

Table 7

The effect of rearing environment on individual feather scores at lay. The table shows the mean and range of continuous variables and the different categories for categorical variables.

Variable		Mean + range/categories	Est (SE)	P-value	N
Constant			18.583 (0.817)	<0.0001	34
	Number of layer sheds on rearing farm	3.2 (0–20)	0.173 (0.077)	0.01	34
Age (days)		254.9 (238–336)	–0.083 (0.017)	<0.0001	34
	Feather pecking started at 16 weeks	No	3.062 (0.864)	0.0002	29
		Yes	Reference category		5

pecks/bird/h. At 16 weeks, SFP was seen in 45% of the 31 flocks, with a mean rate of 0.4 pecks/bird/h for all flocks, or 0.9 pecks/bird/h in the 14 flocks that showed it. The binary logistic regression analysis for the effect of rearing environment on the presence of SFP during rear resulted in seven significant variables (Table 4). The probability of SFP was reduced with a longer duration of the light period, when the main caretaker had more experience with chicks, when a higher percentage of birds were foraging and in houses with a higher minimum ceiling height. The probability of SFP further decreased 64 times when flocks were subjected to one less diet change. It was further reduced in barn and free range compared with organic flocks and in beak trimmed flocks.

At 35 weeks, SFP was observed in 65% of the 34 flocks. The mean rate of SFP was 1.3 pecks/bird/h for all flocks combined or 1.9 pecks/bird/h for the 22 flocks that showed SFP during the visit. The highest observed rate was 10 pecks/bird/h. The rate of SFP at 35 weeks was slightly higher than during rear. Table 5 shows the significant variables for the effect of rearing environment on the probability and rate of SFP at 35 weeks. The rate as well as the probability was analysed here, because the model for the probability of SFP only gave one significant variable. The probability of SFP at 35 weeks was reduced when the range in sound level in the house was less at the end of rear (Table 5, upper half). The rate of SFP was reduced with age and was also associated with feeder type, it was significantly lower when more than one type of feeder was used and significantly higher when compartmentalised pans were used, both compared with the use of chain feeders only. SFP at lay also depended on visit date, it was lowest in summer (Table 5, lower half). Originally the flock that performed 10 pecks/bird/h was removed from the analysis, as its rate was so much higher than the average. However, as the residuals looked normal and the model did not change with the exclusion of the apparent outlier, this data point was retained.

3.3. Percentage of the flock with missing feathers

The percentage of birds with missing feathers was 12% at 16 weeks. Only three of the flocks showed no missing feathers at all, while in another 17 flocks less than 10% of the birds had missing feathers. The remaining 11 flocks contained >10% of birds with missing feathers. The highest percentage was 83%. The percentage of the flock with missing feathers at 16 weeks was reduced when flocks performed less SFP during rear, when a larger area of the rearing house existed of litter area and with a higher percentage of the birds foraging during rear (Table 6, top). Inclusion of the outlying flock (83%) in the model made all these variables insignificant and also provided unsatisfactory residuals. For this reason this flock was removed from the data set to provide a satisfactory model. Following its removal the mean percentage of the flock with missing feathers dropped to 10%. Table 6 (top) therefore presents the model with this outlier excluded.

The mean percentage of birds with missing feathers at 35 weeks was 49%, which was considerably higher than at 16 weeks. Five of the 34 flocks showed no missing feathers at all, while in four flocks 100% of the birds had

missing feathers. Just over half of the flocks (18 of 34) had <50% birds with missing feathers. The percentage of the flock with missing feathers at 35 weeks was reduced when feather pecking had not started at the end of rear, with a lower average sound level at 16 weeks, when more people inspected the birds during rear and in younger birds (Table 6, bottom). Beak trimming status was not significant in the models for both ages, indicating that in this analysis the percentage of birds with missing feathers was unaffected by whether or not the birds were beak trimmed.

3.4. Individual feather scores

For the 34 flocks involved in this study, the mean individual feather score at 35 weeks was 21. Seven flocks had a mean score of 18 or below, with the poorest condition found in three flocks with a mean score of 14. At the other end of the spectrum, eight flocks had a mean perfect score of 24. Individual feather scores improved when feather pecking had not started at 16 weeks, when the rearing farm had a higher number of layer sheds on site and when birds were younger (Table 7). Beak trimming status was also not significant in this model.

3.5. Condition of beak, comb and vent

The lower half of the beak was longer and split in 1.4% of the birds, while in 7.7% of the birds it was longer and not straight. In another 27.3% of birds the lower half was longer and the beaks of the remaining 63.6% showed no abnormalities. Only one bird (0.1%) showed a bloody wound on its comb, while there were only two birds (0.3%) with more than five pecks or scratches. A further 19.1% showed less than five pecks or scratches on the comb and the remaining 80.4% had no injuries on their combs. Only three birds (0.4%) had bloody wounds on their vent, while another three showed five or more pecks/scratches. A further 5.6% showed less than five pecks or scratches and the remaining 93.5% showed no injuries.

4. Discussion

The aim of this study was to investigate the effect of rearing environment on feather pecking in young and adult laying hens. The study confirmed that feather pecking occurs during the rearing period, as SFP was present in almost half of the flocks and approximately 12% of birds in all flocks had missing feathers at the end of the rearing period. It also confirmed that the percentage of the flock with missing feathers at 35 weeks was significantly lower and individual feather scores significantly higher (better) when feather pecking had not started at the end of rear, which is similar to results found by Lambton et al. (2010). Condition of comb and vent was generally good, though just under 40% of beak trimmed birds showed abnormalities of the beak.

The final statistical models presented here were not the only possible models. The presented models included only those variables that made statistical as well as biological sense, but given the large number of possible predictor variables, many of which showed some degree

of correlation, there was inevitably a subjective element to the process of choosing between a number of similar and closely competing models.

In this study it was decided not to use outlying data points (two flocks that did not perform GFP during lay and one flock that had a considerably higher percentage of birds with missing feathers at 16 weeks than average). Including these outlying data points meant that normality assumptions could not be met as shown by the residuals, invalidating the statistical modelling process. Excluding data points is not ideal as these unusual flocks may be able to tell us more about why that particular flock performed so much more or less feather pecking than the rest. The problem was that the data set was not large and having only one or two of such outlying flocks did not provide sufficient information to determine why they were different from the rest in terms of the predictor variables.

The first two outliers were found in the data for GFP at lay. GFP was observed in all but two flocks at lay during the observation periods. It is not clear why these flocks did not perform any GFP during the visit, but it was most likely a coincidence as observations only lasted for 40 min and GFP may not always be observed during this time. The same flocks showed different levels of SFP during lay, in one of them SFP was not observed, whereas in the other flock the rate was above average (4.2 pecks/bird/h). There was also one flock that had a much higher percentage of birds with missing feathers at 16 weeks than the others surveyed. While the average percentage was 12%, 83% of the birds of this particular flock showed missing feathers. Given that it was one flock out of 34 a 95% confidence interval for the prevalence of a flock like this is quite wide at 0.5–15%. This flock had problems with SFP before they were eight weeks old, when bloody wounds were already present on their lower backs. According to the farmer the problem arose during a spell of extremely warm weather during which the birds did not receive any form of ventilation. The house was new and the fans were not yet installed, which may possibly have resulted in heat stress during this particularly warm period. Removing the outliers in this study affected the different models differently, the model for GFP during lay changed only slightly while the one relating to the percentage of the flock with missing feathers changed substantially, showing that the missing feathers model was less robust to changes.

The three models on the effect of rearing environment on feather pecking during rear contained 21 significant variables. Approximately a third of those related to house climate, while another third related to foraging. Foraging itself appeared in all three models, suggesting that it is one of the major factors in reducing feather pecking. The four models on the effect of rearing environment on feather pecking at 35 weeks contained 17 significant variables and sound level was significant in three of the four. The remainder of the discussion is focussed on the main outcomes.

There is little literature on the effect of house climate on feather pecking. Hughes and Duncan (1972) found no effect of brooding temperature on feather damage in 0–20 week old birds. For laying hens, Green et al. (2000) found that feather pecking increased when hens were housed at <20°C, while Lambton et al. (2010) found that GFP

decreased when house temperature increased. It is hard to compare these results, as temperature requirements for laying hens differ from those for chicks. The temperature and humidity levels measured here lay within the industry guidelines (Stonegate Ltd., 2008). Assuming that the industry has set these climatic guidelines at the optimum level, birds seemed to perform most GFP when temperature and humidity lay within their comfort zone. Some propose GFP is explorative behaviour, which comes naturally to the birds and is not harmful, as it does not result in plumage damage (Riedstra and Groothuis, 2002).

The re-occurring effect of climate and especially noise on feather pecking was an interesting though slightly surprising find, as it has not often been identified in previous work apart from Drake et al. (2010). GFP during rear decreased with an increasing range in sound level in the rearing house. If GFP is mainly performed when the birds are comfortable the reduction in GFP might mean that a large range in sound level within the house was aversive. Higher sound levels during rear were further associated with an increased chance of SFP and plumage damage at 35 weeks. Drake et al. (2010) found that flocks that encountered higher sound levels at the end of the rearing period and in the beginning of the laying period, suffered from substantial plumage damage earlier than flocks that were not. They suggested that high sound levels could be a result of feather pecking, rather than its cause, or that noisy birds may be more likely to start feather pecking. This could be true as the birds can make rather loud 'gakel calls' which are considered an indication of frustration. Feather pecking flocks might therefore have higher levels of vocalisation than non-feather pecking flocks (Bright, 2008). However, during the rearing period this is not likely to be the case, as pullets do not yet produce this call. The noise measured here was therefore most likely mechanical noise, caused by fans or brooders. Results of studies testing the link between noise and stress levels, suggest that loud mechanical noise is aversive to the birds (MacKenzie et al., 1993; McAdie et al., 1993). This is potentially problematic, as a good house climate also proved to be important (see below) and as most rearing houses have little or no natural ventilation, fans are needed to maintain a good house climate. If mechanical noise contributes to stress and subsequently to feather pecking, this relationship might pose some problems.

Larger variation in light intensity (within the house, within the visit) was associated with reduced GFP during rear. Kjaer and Vestergaard (1999) found that exploratory GFP was higher in low light intensity (3 versus 30 lx), but as the light intensity within the house was very variable in the current experiment (mean variation within house 49 lx), low light was unlikely to have caused GFP here. Lambton (2008) found increased GFP at 40 weeks in flocks reared without access to daylight. Assuming that rearing with access to daylight increases the difference in light intensity in the house, this would mean that the effect is similar. However, in this analysis the variable 'daylight in house' was not significant in the model so we cannot be sure.

During rear, GFP, SFP and the percentage of the flock with missing feathers were lower when more birds were foraging, which is in agreement with the literature

(reviewed by Rodenburg et al., 2013). Interestingly, the number of birds foraging during rear seemed to be associated only with feather pecking during rear itself, while it never occurred in any of the models relating to the effect of rearing environment on feather pecking at 35 weeks. The effect of foraging therefore seems to be a direct one and does not seem to protect against later feather pecking. Blokhuis and van der Haar (1989), Nicol et al. (2001) and Dixon and Duncan (2010) also found that the current substrate of adult hens was equally or more important than previous experience with litter. The percentage of the flock with missing feathers at 16 weeks further reduced when flocks had a higher percentage of litter area (relative to slats) available to them during rear. The probability of SFP was further higher in organic compared to free range and barn flocks. One of the main differences between the farm types was the percentage of litter area available to the birds. These results are very likely to be directly related to foraging, as a larger litter area enables more birds to forage at the same time, though other factors, like for instance nutrition, might have bearing as well.

Flocks with access to bell drinkers performed the most GFP. Bell drinkers during rear resulted in earlier significant plumage damage at lay (Drake et al., 2010). In adult hens bell drinkers have been found to increase feather pecking (Green et al., 2000; Zimmerman et al., 2006) and vent pecking (Pötzsch et al., 2001). Perhaps spillage from bell drinkers situated above litter reduces litter quality, which would reduce the opportunity to forage. Alternatively, the more equal distribution of nipple drinkers might reduce competition for water. Flocks with access to compartmentalised pan feeders at the end of rear, whose design might also increase competition, severe feather pecked significantly more at 35 weeks than flocks with chains, pans or a mix of feeders. Earlier experiments by Huon et al. (1986) also showed that birds found partitioned feeders less attractive, as shown by shorter durations of feeding bouts. Compartmentalised pans have plastic divisions, which may lead to competition and possible frustration. However, because the number of observations differed for the different types of feeders, this result should be interpreted with caution. That competition can lead to more feather pecking was shown by Lindberg and Nicol (1994) who tested the effect of operant feeders. They found that single birds started defending the operant feeders and found a significant increase in feather pecking compared with birds given access to ad lib feeders.

The probability of SFP during rear increased 64 times with every extra diet change the birds experienced during rear, making it one of the most significant effects of the study. Other epidemiological studies have found a similar effect of diet changes on feather pecking (Green et al., 2000) and vent pecking (Pötzsch et al., 2001) in adult laying hens. Literature suggests that the change from more to less preferred feed causes redirected pecking and an increased activity (Dixon et al., 2006). Chicks that went from regular feed to new, flavoured, feed showed more redirected pecking towards the environment and more general activity, possibly seeking other feed sources after being dissatisfied with the new feed. After changing from non-preferred to preferred diet, no such behavioural changes were detected

(Dixon et al., 2006). Perhaps commercial chicks experience a similar dissatisfaction when they are changed from chick crumbs, which are easy to eat and high in energy level, to possibly less preferred grower meal. Dixon and Nicol (2008) tried to reproduce a commercial diet change for 174 Hy-Line hens of 18 weeks old, which resulted in a significant increase in beak related activity, but not in feather pecking. The lack of feather pecking in the second experiment by Dixon and Nicol (2008) might fit with the results of the current study. There was no effect of diet changes on SFP when modelled by itself, but only in combination with other factors. Perhaps diet changes by themselves are not enough to trigger the behaviour, but may act when combined with other stressors.

In the current study the probability of SFP during rear was reduced when the main caretaker had more years of experience. There is not much literature on the effect of the caretaker on feather pecking, but two other epidemiological studies have tested it for laying hens (Bestman and Wagenaar, 2003) or pullets (Lambton, 2008), though so far it has never been a significant variable. The percentage of birds with missing feathers at 35 weeks further decreased when more people inspected the birds during rear. For adult laying hens, Green et al. (2000) also found that feather pecking increased when the birds were inspected by one person only. These effects might be inter-related. Two people are likely to know more than one and therefore problems would be spotted and fixed earlier than when only one person looks after the birds. Habituating the birds to several people should also make them less susceptible to stress.

Flocks with intact beaks had a higher chance of SFP and performed less GFP than beak trimmed flocks, which is in agreement with Lambton et al. (2010). Staack et al. (2007) on the other hand found in an epidemiological study on 50 farms that the percentage of pullets with missing feathers was significantly higher in beak trimmed flocks (53% versus 30%), though the effect was opposite at lay. They thought this was because beak trimmed flocks were generally kept in conditions which were more likely to cause problems, like higher stocking density and later access to litter and perches.

Interestingly, age was only significant in models relating to the laying period, indicating that none of the outcome variables were associated with age during rear. GFP at 35 weeks significantly decreased, while SFP significantly increased and plumage condition deteriorated with age. This is in agreement with Lambton et al. (2010).

5. Conclusions

This study confirmed the importance of the rearing period in the prevention of feather pecking in laying hen pullets. Some of the major rearing factors associated with feather pecking at both rear and lay were variables relating to house climate and especially sound level. Good foraging reduced feather pecking during rear. Not only did foraging itself reduce problems with feather pecking during rear, the availability of a large litter area with friable litter also proved important. Foraging during rear was not related to feather pecking at 35 weeks. The third interesting

outcome was the effect of diet changes during rear on feather pecking during rear and lay, which was also suggested in previous work. Lastly, this study came up with several indications that greater experience amongst rearing staff was protective against feather pecking during rear and at 35 weeks.

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